

$$\begin{aligned}
 1. \quad 0101 &= 1 \times 2^3 + 1 \times 2^2 = 5 \\
 0110 &= 1 \times 2^3 + 1 \times 2^2 = 6 \\
 1001 &= 1 \times 2^3 + 1 \times 2^0 = 9
 \end{aligned}$$

Decimal = (569)

$$2. \quad (52)_{10} = 00110100$$

$$(23)_{10} = 00010111$$

$$2's \text{ complement of } (23)_{10} = 11101000 + 1 = 11101001$$

$$(52)_{10} - (23)_{10} = 00110100$$

$$11101001$$

$$\textcircled{1}00011101$$

carry is ①
hence positive

$$\text{Ans } 00011101 \text{ is } 1 + 4 + 8 + 16 = 29$$

$$(23)_{10} = 00010111$$

$$(52)_{10} = 00110100$$

$$2's \text{ complement} = 11001011 + 1 = 11001100$$

$$\begin{array}{r}
 (23)_{10} - (52)_{10} = 00010111 \\
 \quad \quad \quad 00110100 \\
 \hline
 01001011
 \end{array}$$

$$\begin{array}{r}
 00010111 \\
 11001100 \\
 \hline
 11100011
 \end{array}$$

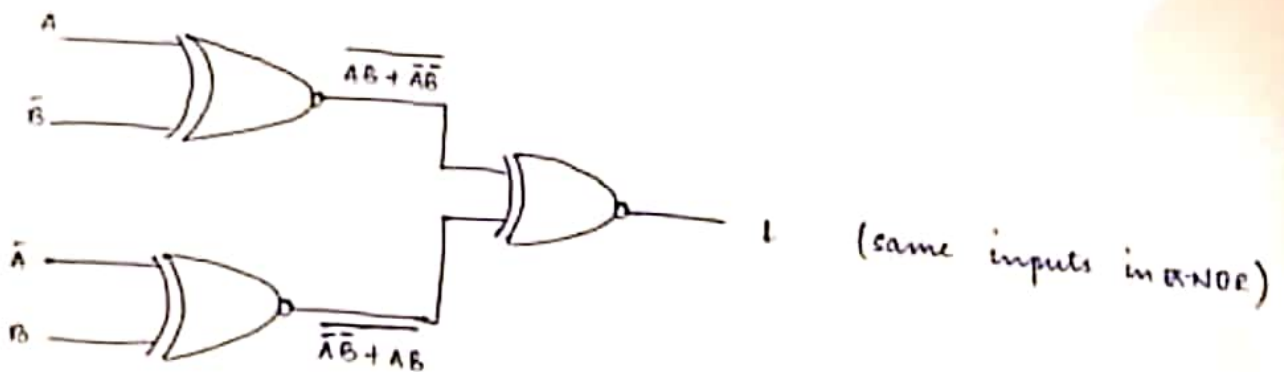
$$\begin{aligned}
 &= 2^2 + 2^1 + 2^5 + 2^4 + 2^0 \\
 &= 64 + 32 + 2 + 1 + 128 = 227
 \end{aligned}$$

Ans is negative and 2's complement of 227

$$\begin{aligned}
 2's \text{ complement} &= 00011101 = 2^4 + 2^3 + 2^2 + 1 = 16 + 8 + 4 + 1 \\
 &= -29
 \end{aligned}$$

$$\text{Ans: } -29$$

3.



$$4. \quad ABC + ABC + ABC + ABC = ABC$$

5.

AB \ CD	00	01	11	10
00			1	1
01				
11	1	1	1	1
10			1	1

$$Y = \underline{I} + \underline{II} + \underline{III}$$

$$= AC + \bar{B}C + AB$$

6.

AB \ CD	00	01	11	10
00	0			
01				0
11	0			
10				0

$$F(A, B, C, D) = \sum M(0, 6, 10, 12)$$

$$Y = (A+B+C+D) \cdot (\bar{A}+\bar{B}+C+D) \cdot (A+\bar{B}+\bar{C}+D) \cdot (\bar{A}+B+\bar{C}+D)$$

7.

AB \ CD	00	01	11	10
00	1	0	1	1
01	1	1	0	0
11	0	0	0	0
10	1	1	0	1

$$Y = \bar{A}\bar{C}\bar{D} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + \bar{B}C\bar{D}$$

8. $\bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + ABC = A + B\bar{C}$

9.

A	B	Cin	S	Co
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Sum

BC	00	01	11	10
0		1		1
1	1		1	

$$S = A\bar{B}\bar{C} + \bar{A}\bar{B}C + ABC + \bar{A}B\bar{C}$$

$$= A \oplus B \oplus C_{in}$$

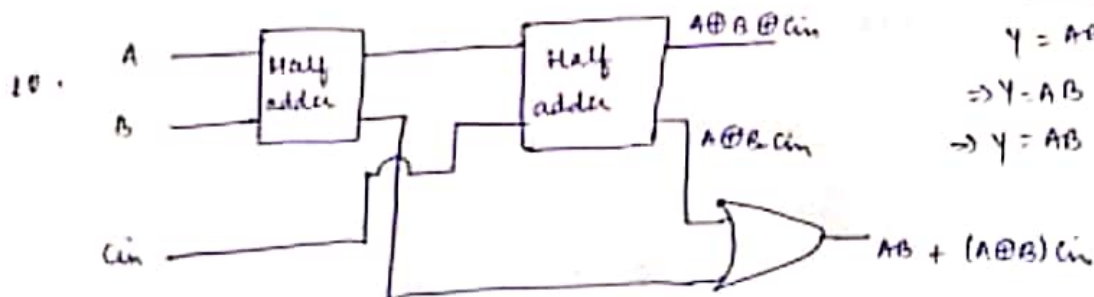
Carry

BC	00	01	11	10
0			1	
1		1	1	1

$$Y = AB + A\bar{B}C_{in} + \bar{A}B C_{in}$$

$$\Rightarrow Y = AB + C_{in}(A\bar{B} + \bar{A}B)$$

$$\Rightarrow Y = AB + (A \oplus B) \cdot C_{in}$$



Sum of full adder $A \oplus B \oplus C$
 Sum of half adder $A \oplus B$
 So
 Sum of two half adder with third output in second adder gives sum in full adder

similarly when carry of first half adder and second half adder are fed to OR gate it gives carry of full adder.